Think — choosing a chart

Four kinds of diagram, four kinds of question — and the charts that quietly mislead.

LESSON 168 | 1 × 40 MIN | MATEMATIKA 6, O‘YLAB KO‘R | S7 16 INTERPRETING RESULTS

LESSON CLOCK

10' | 12' | 12' | 6'

Four charts, four questions	10 min
Choosing, with a reason	12 min
Charts that mislead	12 min
The task	6 min

LEARNING OBJECTIVES

- Match each kind of data to the chart that shows it best.
- Justify the choice in one sentence.
- Recognise a truncated axis and other misleading devices.
- State the three things every chart must carry.

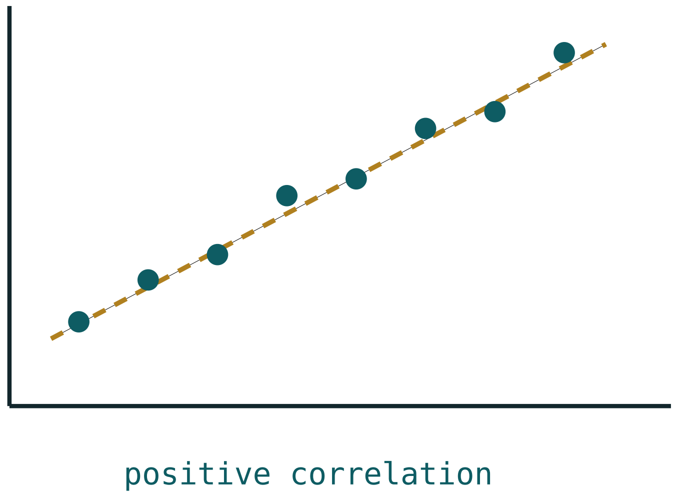
TEXTBOOK

National	pp. 480–482
Cambridge	Stage 7, pp. 184–188
Workbook	Investigation 16

01 Explanation

Four charts, four questions

CHART	ANSWERS THE QUESTION	GOOD EXAMPLE
bar chart	how do the categories compare in size?	pupils in each of six grades
pie chart	how is one whole shared out?	the school budget by item
line graph	how has one quantity changed over time?	the noon temperature each month
scatter graph	are two measurements related?	height against arm span



A scatter graph — the only one of the four that shows a relationship between two measurements

ASK WHAT THE READER SHOULD SEE AT A GLANCE

A bar chart makes sizes comparable; a pie chart makes shares obvious; a line graph makes a trend visible; a scatter graph makes a relationship visible. Choose the one whose strength is the question.

Choosing, with a reason

THE DATA	BEST CHART	WHY
favourite sport of 60 pupils	pie chart	the 60 form one whole to be shared
pupils in each of six grades	bar chart	six sizes to be compared, not shares
rainfall each month for a year	line graph	the months are in order and the trend matters
height against arm span	scatter graph	two measurements from each pupil
marks out of ten for one class	bar chart of the frequencies	the marks are ordered categories

A LINE GRAPH NEEDS AN ORDERED HORIZONTAL AXIS

Joining “football” to “tennis” with a line says that something exists between them. Nothing does. Time, length and temperature may be joined; favourite sports may not.

Charts that mislead

DEVICE	WHAT THE READER THINKS	THE TRUTH
the vertical axis starts at *40*, not *0*	one bar is three times another	*44* against *52* — a small difference
bars of different widths	the wider bar means more	only the height carries the number
a pie chart with *15* sectors	nothing at all	the thin sectors cannot be read
two pie charts, no totals given	the groups are equal	shares are equal; the numbers may not be
a pictogram with pictures of different sizes	a big picture means many	each picture should stand for a fixed number

> THREE THINGS EVERY CHART MUST CARRY
>
> A title that says what is shown; labelled axes or a key; and the total number the data came from. A chart missing any of the three can be read in more than one way.

The task

> TODAY'S TASK
>
> For each of these five sets of data, name the chart you would draw and give one sentence of reason — then say what would be lost if the wrong one were chosen instead.

DATA	WRONG CHOICE TO ARGUE AGAINST
the *60* pupils' favourite sports	a line graph
the noon temperature each month	a pie chart
height and arm span of *30* pupils	a bar chart
the population of six grades	a pie chart with no total
a class's marks out of ten	a pie chart with *11* sectors

02 Worked examples

EXAMPLE 1 **Sixty pupils name a favourite sport. Which chart, and why?**

1. The *60* pupils make one whole that is being shared out.
2. A pie chart shows shares directly.
3. A bar chart would also work if the question is “which is largest”.
   A line graph would not: sports have no order.

ANSWER

A pie chart — the data is shares of one whole

EXAMPLE 2 **Monthly rainfall for a year. Which chart, and why?**

1. The months are in order and the interest is the trend.
2. A line graph joins them meaningfully.
3. A pie chart would hide the order entirely.
   Which month was wettest would be almost unreadable.

ANSWER

A line graph — ordered time on the horizontal axis

EXAMPLE 3 **Two bars stand for *44* and *52*, drawn on an axis starting at *40*. Why does this mislead?**

1. The drawn heights are 4 and 12 — a ratio of 1 to 3.
2. The real values differ by under a fifth.
3. Starting the axis at 0 shows the truth.

   Or mark the break in the axis clearly.

ANSWER

The heights exaggerate the difference by a factor of about three

03 Interactive model

Draw the same two bars twice on the board — once from zero and once from forty. The class sees the trick before it is named.

Which chart, and why

Shares of one whole are best shown by:

- a bar chart
- a pie chart
- a line graph
- a scatter graph

Change over time is best shown by:

- a pie chart
- a line graph
- a scatter graph
- a pictogram

A relationship between two measurements needs:

- a bar chart
- a pie chart
- a scatter graph
- a line graph

Joining “football” to “tennis” with a line is wrong because:

- the colours differ
- the categories have no order
- the totals differ
- lines are hard to draw

An axis starting at 40 rather than 0:

- saves space honestly
- exaggerates differences
- is always wrong
- changes the data

A pie chart with 15 sectors is:

- clearer
- unreadable
- more accurate
- preferred

Bars of different widths suggest:

- nothing
- that width carries meaning
- a larger total
- better data

Every chart must carry:

- a title, labels and the total
- colour
- a percentage
- a protractor

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Bar chart	Ustunli diagramma	Столбчатая диаграмма
Pie chart	Doiraviy diagramma	Круговая диаграмма
Line graph	Chiziqli grafik	Линейный график
Scatter graph	Sochilish diagrammasi	Диаграмма рассеяния
Category	Toifa	Категория
Axis	O‘q	Ось
Misleading	Chalg‘ituvchi	Вводящий в заблуждение
Title and key	Sarlavha va izoh	Заголовок и легенда

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A pie chart is best for:

- a trend
- shares of a whole
- two measurements
- a timetable

A bar chart compares:

- shares
- sizes across categories
- time only
- nothing

A scatter graph needs:

- one value each
- two values from each person
- a total
- angles

A truncated vertical axis:

- is always fine
- exaggerates differences
- shrinks differences
- has no effect

Two pie charts without totals compare:

- numbers
- shares only
- nothing
- means

A pictogram's pictures must:

- vary in size
- each stand for a fixed number
- be coloured
- be circles

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The chart for shares of one whole

   ANSWER A pie chart

2. The chart for comparing counts across categories

   ANSWER A bar chart

3. The chart for change over time

   ANSWER A line graph

4. The chart for a relationship between two measurements

   ANSWER A scatter graph

5. The sectors of a pie chart total

   ANSWER $360°$

6. Bars in a bar chart should have

   ANSWER Equal widths, with gaps

7. Every chart needs a title, labels and

   ANSWER The total the data came from

Medium · the standard question 7 PROBLEMS

1. Favourite sports of *60* pupils

   ANSWER A pie chart (or a bar chart)

2. The noon temperature for twelve months

   ANSWER A line graph

3. Height against arm span for *30* pupils

   ANSWER A scatter graph

4. The number of pupils in each of six grades

   ANSWER A bar chart

5. The share of the school budget by item

   ANSWER A pie chart

6. A bar chart whose axis starts at *40*

   ANSWER Misleading

7. Two pie charts of different totals compare

   ANSWER Shares, not numbers

Hard · stretch and reason

7 PROBLEMS

1. Why is a pie chart poor for 15 categories?

   ANSWER The sectors are too thin to read or label

2. Why is a line graph wrong for favourite sports?

   ANSWER The categories have no order

3. Bars of different widths suggest

   ANSWER That the area carries meaning, which it does not

4. The best chart for one class's marks out of ten

   ANSWER A bar chart of the frequencies

5. The best chart for "do taller pupils have longer arms?"

   ANSWER A scatter graph

6. A pie chart of 7 people needs angles to

   ANSWER 1 decimal place, adjusted to total $360°$

7. Bars of 44 and 52 on an axis from 40

   ANSWER Look like 1 to 3

07 Homework

Homework — the task

One page: five choices, five reasons, and one sentence on what the wrong chart would hide.

1. Name the chart you would draw for each of the five data sets in the last section.
2. Give one sentence of reason for each choice.
3. For each, say what would be lost if the wrong chart named beside it were used.
4. Draw the two-bar example twice — from zero and from forty — and label which is honest.
5. List the three things every chart must carry.